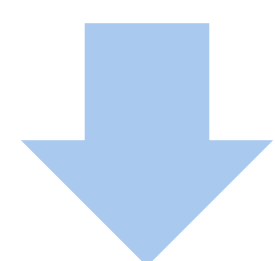
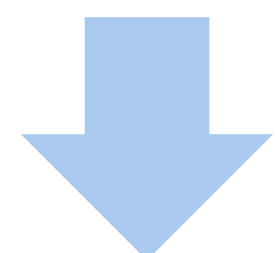


BUILD

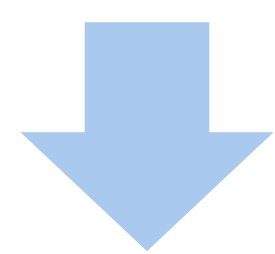
Discharge notes
328,839 MIMIC-IV



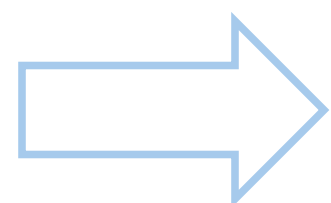
Qwen extraction
cause → effect triples



UMLS normalization
merge synonyms to CUIs



Causal KG
4.3K concepts / 19K edges



AUDIT

Which validation signals deserve trust?

Scored against SemMedDB literature causal predicates as an independent reference.

WORKS

Co-occurrence lift

existence validation
removes spurious co-mentions

FAILS

Coding-order time

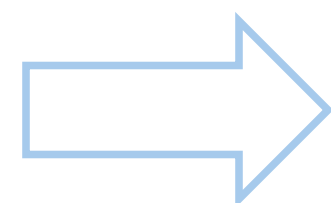
direction validation
54.2% vs 73.6%; coding time != onset

PARTIAL

Literature corroboration

direction + confidence
74% agreement, ~30% edge coverage

**Takeaway: existence is easy to validate;
direction remains uncertain.**

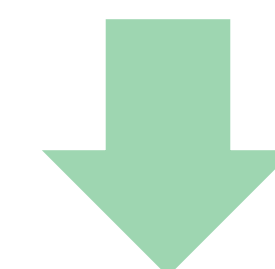


APPLY

**Lift-validated
causal KG**



**Causal RAG
retriever + Qwen**



**Held-out patient
causal question**

**+7.9 pts
vs unvalidated graph**